

16 January 2026

Dr. Andrea Fuller
Editor-in-Chief
Conservation Physiology

Dear Dr. Fuller and Editorial Board,

Please find enclosed a manuscript titled “Xenobiotic Exposure May Confound Traditional Stress Biomarkers: Evaluating PFAS and Cadmium Effects on Largemouth Bass (*Micropterus nigricans*) Leukocyte Profiles” by Johnson *et al.* for consideration as a Research Article in *Conservation Physiology*. The manuscript is original, not under review elsewhere, and all authors have approved its submission.

The authors believe that this manuscript will be highly relevant to the journal's audience. Our study examined how exposure to cadmium (Cd) and perfluoropentanoic acid (PFPeA), a short-chain per- and polyfluoroalkyl substance (PFAS), affects a common chronic stress biomarker—the neutrophil-to-lymphocyte ratio (NLR)—in largemouth bass under controlled laboratory conditions. Using a replicated tank experiment with multiple contaminant levels and repeated blood sampling, we tested whether xenobiotic exposure influences baseline stress, alters responses to an abrupt salinity increase, and impairs post-stressor recovery. High PFPeA exposure elevated baseline NLR, dampened the acute response to salinity, and prevented recovery one week after the stressor, whereas Cd and low PFPeA exposure did not differ from controls.

Our findings demonstrate that short-chain PFAS can substantially alter vertebrate stress physiology and suggest that unmeasured contaminant exposure may confound interpretations of traditional stress biomarkers in natural systems. Given the ecological and socioeconomical importance of largemouth bass and the widespread use of leukocyte profiles in conservation physiology, this work has broad relevance for researchers and managers assessing organismal health in contaminated and changing aquatic environments.

The manuscript is formatted as a Research Article and fully complies with the journal's guidelines. The study was approved by the Southeastern Louisiana University IACUC (protocol 0093), and all required statements are included.

We thank you for your time and consideration. I would be happy to provide any additional information if needed.

Sincerely,

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